Appendix C: Research Hypotheses and Experimental Recommendations

C.1 Falsifiable Research Hypotheses

C.1.1 RVI-01: Reproducible Benchmarking

Hypothesis 1.1: Environmental Variation Effects

Statement: Autonomous visual inspection performance degrades systematically under controlled environmental variations (lighting, object placement, occlusion), with degradation magnitude varying across robot platforms and planning methods.

Independent variables:

- Lighting intensity: 100 lux, 500 lux, 1000 lux, 2000 lux
- Object placement: Position (±10 cm), orientation (±15°)
- Occlusion level: 0%, 25%, 50%, 75% of target surface occluded

Dependent variables:

- Coverage completeness: Percentage of target surface observed above resolution threshold (%)
- Acquisition quality score: Composite metric combining blur, exposure, occlusion, framing (0–1 scale)
- Execution time: Total time from start to completion (seconds)

Controls:

- Fixed robot platform (e.g., Franka Panda manipulator)
- Fixed planning method (e.g., coverage-first viewpoint planning)
- Randomized trial order to control for learning effects
- Fixed target objects (e.g., YCB objects)

Validation:

- Repeated trials: $n \geq 30$ per condition (4 lighting × 3 placement × 4 occlusion = 48 conditions × 30 trials = 1,440 trials)
- Statistical analysis: Three-way repeated measures ANOVA with post-hoc Tukey HSD comparisons
- Effect size: Partial eta-squared ($\eta^2 p$) for each main effect and interaction
- Power analysis: Ensure 80% power to detect medium effect sizes ($\eta^2 p \geq 0.06$)

Expected outcome: Significant main effects of lighting ($F(3,87) > 3.0$, $p < 0.05$), placement ($F(2,58) > 3.0$, $p < 0.05$), and occlusion ($F(3,87) > 3.0$, $p < 0.05$) on coverage completeness and quality score; significant interactions between factors

Falsification criteria: If no significant main effects or interactions are observed ($p > 0.05$), the hypothesis is rejected

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Hypothesis 1.2: Platform Performance Trade-offs

Statement: Humanoid robots exhibit different performance trade-offs compared to manipulators and wheeled robots for visual inspection tasks, with humanoids offering superior workspace coverage but longer execution times and lower repeatability.

Independent variables:

- Robot platform: Humanoid (Unitree G1), manipulator (Franka Panda), wheeled mobile robot (Clearpath Husky with UR5)

Dependent variables:

- Workspace coverage: Percentage of workspace volume accessible for inspection (%)
- Execution time: Total time from start to completion (seconds)
- Pose repeatability: Standard deviation of camera pose across repeated trials (mm for position, degrees for orientation)

Controls:

- Fixed task: Inspect standardized object (e.g., YCB object) from multiple viewpoints
- Fixed environment: Controlled laboratory setup with standardized lighting and object placement
- Fixed planning method: Same viewpoint planning algorithm adapted to each platform

Validation:

- Repeated trials: $n \geq 30$ per platform (3 platforms × 30 trials = 90 trials)
- Statistical analysis: One-way repeated measures ANOVA with post-hoc Tukey HSD comparisons
- Effect size: Partial eta-squared ($\eta^2 p$) for platform effect
- Power analysis: Ensure 80% power to detect medium effect sizes ($\eta^2 p \geq 0.06$)

Expected outcome: Significant main effect of platform on workspace coverage ($F(2,58) > 3.0$, $p < 0.05$), execution time ($F(2,58) > 3.0$, $p < 0.05$), and pose repeatability ($F(2,58) > 3.0$, $p < 0.05$); humanoid > manipulator > wheeled robot for workspace coverage; humanoid > wheeled robot > manipulator for execution time; manipulator > wheeled robot > humanoid for pose repeatability

Falsification criteria: If no significant platform effect is observed ($p > 0.05$) or if the expected ordering is not confirmed, the hypothesis is rejected

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C.1.2 RVI-03: Viewpoint and Motion Planning

Hypothesis 3.1: Humanoid-Specific Planning Effectiveness

Statement: Humanoid-specific viewpoint planning incorporating whole-body kinematics and bipedal stability constraints achieves higher coverage completeness and kinematic feasibility compared to manipulator-centric planning adapted to humanoid platforms.

Independent variables:

- Planning method: Humanoid-specific planning (incorporating whole-body kinematics and stability constraints), adapted manipulator planning (treating humanoid as fixed-base manipulator)

Dependent variables:

- Coverage completeness: Percentage of target surface observed above resolution threshold (%)
- Kinematic feasibility: Percentage of planned viewpoints executable without violating joint limits or stability constraints (%)

- Execution success rate: Percentage of trials completed without failures (%)

Controls:

- Fixed robot platform: Unitree G1 humanoid robot
- Fixed environment: Controlled laboratory setup with standardized lighting and object placement
- Fixed target objects: YCB objects with varying geometry and size

Validation:

- Repeated trials: n≥30 per method (2 methods × 30 trials = 60 trials)
- Statistical analysis: Paired t-tests for each dependent variable
- Effect size: Cohen's d for each comparison
- Power analysis: Ensure 80% power to detect medium effect sizes ($d \geq 0.5$)

Expected outcome: Humanoid-specific planning > adapted manipulator planning for coverage completeness ($t(29) > 2.0$, $p < 0.05$, $d > 0.5$), kinematic feasibility ($t(29) > 2.0$, $p < 0.05$, $d > 0.5$), and execution success rate ($t(29) > 2.0$, $p < 0.05$, $d > 0.5$)

Falsification criteria: If no significant differences are observed ($p > 0.05$) or if adapted manipulator planning outperforms humanoid-specific planning, the hypothesis is rejected

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Hypothesis 3.2: Joint Optimization Benefits

Statement: Joint optimization of viewpoint selection and motion planning outperforms sequential planning (viewpoint selection followed by motion planning) in terms of coverage completeness, execution time, and motion smoothness.

Independent variables:

- Planning approach: Joint optimization (simultaneous viewpoint and motion planning), sequential planning (viewpoint selection followed by motion planning)

Dependent variables:

- Coverage completeness: Percentage of target surface observed above resolution threshold (%)
- Execution time: Total time from start to completion (seconds)
- Motion smoothness: Root mean square jerk (m/s^3) and acceleration (m/s^2)

Controls:

- Fixed robot platform: Franka Panda manipulator or Unitree G1 humanoid robot
- Fixed environment: Controlled laboratory setup with standardized lighting and object placement
- Fixed target objects: YCB objects with varying geometry and size

Validation:

- Repeated trials: n≥30 per approach (2 approaches × 30 trials = 60 trials)
- Statistical analysis: Paired t-tests for each dependent variable
- Effect size: Cohen's d for each comparison
- Power analysis: Ensure 80% power to detect medium effect sizes ($d \geq 0.5$)

Expected outcome: Joint optimization > sequential planning for coverage completeness ($t(29) > 2.0$, $p < 0.05$, $d > 0.5$) and motion smoothness ($t(29) > 2.0$, $p < 0.05$, $d > 0.5$); joint optimization < sequential planning for execution time ($t(29) > 2.0$, $p < 0.05$, $d > 0.5$)

Falsification criteria: If no significant differences are observed ($p > 0.05$) or if sequential planning outperforms joint optimization on coverage or smoothness, the hypothesis is rejected

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C.1.3 RVI-06: Acquisition Quality Control

Hypothesis 6.1: Quality-Aware NBV Effectiveness

Statement: Quality-aware next-best-view planning that integrates image quality assessment into viewpoint selection achieves higher acquisition quality and fewer re-observations compared to coverage-first planning with post-hoc quality filtering.

Independent variables:

- Planning method: Quality-aware NBV (integrates quality assessment into viewpoint selection), coverage-first with filtering (selects viewpoints for coverage, filters poor-quality images, triggers re-observation)

Dependent variables:

- Acquisition quality score: Composite metric combining blur, exposure, occlusion, framing (0–1 scale)
- Number of re-observations: Total number of viewpoints requiring re-observation due to poor quality
- Execution time: Total time from start to completion (seconds)

Controls:

- Fixed robot platform: Franka Panda manipulator or Unitree G1 humanoid robot
- Fixed environment: Controlled laboratory setup with varying lighting conditions
- Fixed target objects: YCB objects with varying geometry and reflectance

Validation:

- Repeated trials: $n \geq 30$ per method (2 methods × 30 trials = 60 trials)
- Statistical analysis: Paired t-tests for each dependent variable
- Effect size: Cohen's d for each comparison
- Power analysis: Ensure 80% power to detect medium effect sizes ($d \geq 0.5$)

Expected outcome: Quality-aware NBV > coverage-first with filtering for acquisition quality score ($t(29) > 2.0$, $p < 0.05$, $d > 0.5$); quality-aware NBV < coverage-first with filtering for number of re-observations ($t(29) > 2.0$, $p < 0.05$, $d > 0.5$); quality-aware NBV ≈ coverage-first with filtering for execution time ($t(29) < 2.0$, $p > 0.05$)

Falsification criteria: If no significant differences are observed for quality score or re-observations ($p > 0.05$), or if coverage-first with filtering outperforms quality-aware NBV, the hypothesis is rejected

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Hypothesis 6.2: Proactive vs. Reactive Quality Control

Statement: Proactive quality-aware viewpoint planning reduces the frequency of poor-quality acquisitions (blur, over/underexposure, occlusion) compared to reactive quality control that triggers re-observation after detecting poor quality.

Independent variables:

- Quality control strategy: Proactive planning (selects viewpoints to maximize expected quality), reactive re-observation (detects poor quality and triggers re-observation)

Dependent variables:

- Frequency of poor-quality acquisitions: Percentage of viewpoints with quality score < 0.5 (%)
- Number of re-observations: Total number of viewpoints requiring re-observation
- Execution time: Total time from start to completion (seconds)

Controls:

- Fixed robot platform: Franka Panda manipulator or Unitree G1 humanoid robot
- Fixed environment: Controlled laboratory setup with varying lighting conditions
- Fixed target objects: YCB objects with varying geometry and reflectance

Validation:

- Repeated trials: n≥30 per strategy (2 strategies × 30 trials = 60 trials)
- Statistical analysis: Paired t-tests for each dependent variable
- Effect size: Cohen's d for each comparison
- Power analysis: Ensure 80% power to detect medium effect sizes ($d \geq 0.5$)

Expected outcome: Proactive planning < reactive re-observation for frequency of poor-quality acquisitions ($t(29) > 2.0$, $p < 0.05$, $d > 0.5$) and number of re-observations ($t(29) > 2.0$, $p < 0.05$, $d > 0.5$); proactive planning ≈ reactive re-observation for execution time ($t(29) < 2.0$, $p > 0.05$)

Falsification criteria: If no significant differences are observed for poor-quality frequency or re-observations ($p > 0.05$), or if reactive re-observation outperforms proactive planning, the hypothesis is rejected

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C.1.4 RVI-07: Simulation-to-Reality Transfer

Hypothesis 7.1: Visibility Model Fidelity

Statement: Simulated visibility models exhibit systematic bias compared to physical measurements, with bias magnitude varying across occlusion types (self-occlusion, external occlusion, sensor limitations) and calibration methods.

Independent variables:

- Occlusion type: Self-occlusion (object geometry), external occlusion (obstacles), sensor limitations (field-of-view, resolution)
- Calibration method: No calibration (default simulator parameters), geometric calibration (camera intrinsics/extrinsics), full calibration (geometric + sensor noise + illumination)

Dependent variables:

- Visibility prediction error: Absolute difference between simulated and measured visibility

(percentage of surface visible)
- Coverage prediction error: Absolute difference between simulated and measured coverage (percentage of surface observed above resolution threshold)
- Occlusion detection error: False positive and false negative rates for occlusion detection

Controls:

- Fixed robot platform: Franka Panda manipulator with Intel RealSense D435i camera
- Fixed environment: Controlled laboratory setup with standardized lighting and object placement
- Fixed target objects: YCB objects with known geometry
- Independent ground truth: Laser tracker or photogrammetry system

Validation:

- Repeated trials: n≥30 per condition (3 occlusion types × 3 calibration methods = 9 conditions × 30 trials = 270 trials)
- Statistical analysis: Two-way repeated measures ANOVA with post-hoc Tukey HSD comparisons
- Effect size: Partial eta-squared ($\eta^2 p$) for each main effect and interaction
- Power analysis: Ensure 80% power to detect medium effect sizes ($\eta^2 p \geq 0.06$)

Expected outcome: Significant main effects of occlusion type ($F(2,58) > 3.0$, $p < 0.05$) and calibration method ($F(2,58) > 3.0$, $p < 0.05$) on visibility and coverage prediction errors; full calibration < geometric calibration < no calibration for prediction errors

Falsification criteria: If no significant main effects or interactions are observed ($p > 0.05$), or if calibration does not reduce prediction errors, the hypothesis is rejected

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C.2 Experimental Recommendations

C.2.1 Datasets and Benchmark Environments

Recommended Datasets

1. YCB Object and Model Set

- Description: Standardized set of 77 objects with known geometry, texture, and physical properties
- Relevance: Enables coverage evaluation with ground truth geometry
- Usage: Target objects for inspection tasks; coverage completeness validation
- Availability: http://www.ycbbenchmarks.com/

2. ShapeNet

- Description: Large-scale 3D object dataset with 51,300 models across 55 categories
- Relevance: Enables large-scale simulation experiments with diverse object geometries
- Usage: Simulation-based evaluation; generalization testing
- Availability: https://shapenet.org/

3. Replica

- Description: Photo-realistic 3D scene models for indoor environments
- Relevance: Enables realistic simulation of navigation and inspection tasks

- Usage: Simulation-based evaluation; sim-to-real transfer validation
- Availability: https://github.com/facebookresearch/Replica-Dataset

4. LIBERO

- Description: Manipulation benchmark with diverse tasks and environments
- Relevance: Enables standardized evaluation of manipulation and inspection tasks
- Usage: Benchmarking; comparison against existing methods
- Availability: https://libero-project.github.io/

5. Custom ARC-RVI Benchmark Dataset

- Description: Curated dataset for autonomous visual inspection evaluation
- Contents:
  - Standardized inspection tasks (object documentation, surface coverage, defect localization)
  - Controlled environmental variations (lighting, object placement, occlusion)
  - Multi-platform validation data (manipulator, mobile robot, humanoid)
  - Ground truth measurements (laser tracker, photogrammetry)
- Relevance: Addresses gap in inspection-specific benchmarking
- Usage: Primary evaluation dataset for ARC-RVI research
- Availability: To be released publicly upon completion

Recommended Benchmark Environments

1. Controlled Laboratory Setup

- Description: Adjustable lighting, object placement, and occlusion for systematic evaluation
- Components:
  - Programmable LED lighting array (100–2000 lux, adjustable color temperature)
  - Precision positioning stage for object placement (±0.1 mm accuracy)
  - Modular occlusion panels (0%, 25%, 50%, 75% occlusion levels)
  - Independent measurement system (laser tracker or photogrammetry)
- Relevance: Enables controlled experimental validation with ground truth
- Usage: Primary physical validation environment

2. Simulated Environments

- Isaac Sim: NVIDIA's physics-based simulation platform with photo-realistic rendering
  - Relevance: Enables large-scale evaluation and domain randomization
  - Usage: Sim-to-real transfer experiments; large-scale benchmarking
  - Availability: https://developer.nvidia.com/isaac-sim
- Gazebo: Open-source robot simulation platform with ROS integration
  - Relevance: Widely used in robotics research; supports diverse robot platforms
  - Usage: Simulation-based evaluation; comparison against existing methods
  - Availability: http://gazebosim.org/
- MuJoCo: Physics engine for robotics and biomechanics simulation
  - Relevance: Fast, accurate physics simulation; supports contact-rich tasks
  - Usage: Simulation-based evaluation; reinforcement learning experiments
  - Availability: https://mujoco.org/

3. Real-World Scenarios

- Office environment: Desks, shelves, cabinets for object inspection
- Warehouse environment: Pallets, boxes, industrial equipment for large-scale inspection
- Manufacturing environment: Assembly lines, workstations, quality control stations

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C.2.2 Baseline Algorithms

Viewpoint Planning Baselines

1. Coverage-First Methods

- Set Covering Problem (SCP): Formulates viewpoint selection as SCP; minimizes number of viewpoints for complete coverage 1 , 2 , 3
- Traveling Salesman Problem (TSP): Formulates viewpoint sequencing as TSP; minimizes travel distance 1 , 2 , 10
- Generalized TSP (GTSP): Formulates scan path selection as GTSP; incorporates kinematic constraints 2

2. Information-Theoretic NBV

- Entropy Reduction: Selects viewpoints maximizing expected entropy reduction in occupancy map 2 , 3 , 5
- Mutual Information: Selects viewpoints maximizing mutual information between observations and scene geometry 2 , 3
- Coverage Gain: Selects viewpoints maximizing expected coverage gain 1 , 5

3. Learning-Based NBV

- 3D CNN-Based Scoring: Uses 3D convolutional neural networks to predict viewpoint utility from occupancy maps 22
- Reinforcement Learning: Trains policies via RL to select viewpoints maximizing cumulative reward 14
- Transformer-Based Generation: Uses Transformer architectures to generate viewpoint sequences 9

Quality Control Baselines

1. Adaptive Exposure Control

- Gradient-Based: Maximizes image gradient information 3 , 16
- Entropy-Based: Maximizes image entropy 6 , 8
- Bayesian Optimization: Uses Bayesian optimization to learn optimal exposure parameters 15 , 23
- Reinforcement Learning: Trains policies via RL to adjust exposure in dynamic lighting 9

2. Noise-Aware Quality Assessment

- Gradient + Entropy + Noise: Combines gradient, entropy, and noise metrics 8
- No-Reference Quality Estimation: Uses learned models to predict quality without reference images 14

3. Motion Blur Detection

- Edge Detection: Uses improved edge detection with Gaussian noise reduction 7

- Laplacian Variance: Uses Laplacian variance to detect blur 27
- IMU-Based Detection: Uses IMU signals to detect motion blur 30

Sim-to-Real Transfer Baselines

1. Domain Randomization

- Visual Randomization: Varies textures, lighting, colors, camera parameters 1 , 2 , 9 , 14
- Physics Randomization: Varies object masses, friction coefficients, contact parameters 1 , 4
- Sensor Randomization: Varies sensor noise, resolution, field-of-view 8 , 14

2. Domain Adaptation

- Feature Alignment: Aligns feature distributions between simulation and reality 10 , 11 , 22
- Adversarial Training: Uses adversarial loss to encourage domain-invariant features 11 , 22
- Dual VAE: Uses dual variational autoencoders with shared decoder 11

3. Real-to-Sim Calibration

- System Identification: Estimates physical parameters from video observations 6 , 17 , 23
- VLM-Guided Calibration: Uses vision-language models to diagnose sim-real mismatches 17 , 23
- 3D Gaussian Splatting: Uses 3D Gaussian splatting to model real scenes 25

C.2.3 Statistical Methods and Validation Procedures

Recommended Statistical Methods

1. Repeated Measures ANOVA

- Purpose: Compare methods across multiple conditions with repeated trials
- Assumptions: Normality, sphericity (Mauchly's test), homogeneity of variance
- Post-hoc tests: Tukey HSD or Bonferroni correction for pairwise comparisons
- Effect size: Partial eta-squared ($\eta^2 p$) to quantify practical significance
- Software: R (ez package), Python (statsmodels), SPSS

2. Paired t-tests

- Purpose: Compare two methods on the same subjects/trials
- Assumptions: Normality of differences, independence of observations
- Effect size: Cohen's d to quantify practical significance
- Software: R (t.test), Python (scipy.stats), SPSS

3. Power Analysis

- Purpose: Determine required sample sizes for detecting meaningful differences
- Parameters: Desired power (typically 0.80), significance level (typically 0.05), expected effect size
- Software: G*Power, R (pwr package), Python (statsmodels)

4. Confidence Intervals

- Purpose: Report uncertainty in effect estimates
- Recommendation: Report 95% CIs for all effect estimates
- Interpretation: If CI excludes zero, effect is statistically significant at $\alpha = 0.05$

Validation Procedures

1. Cross-Validation

- Purpose: Assess generalization of learning-based methods
- Methods: K-fold cross-validation (k=5 or k=10), leave-one-out cross-validation
- Application: Learning-based NBV, quality assessment, sim-to-real transfer

2. Held-Out Test Sets

- Purpose: Independent evaluation data not used during training or tuning
- Recommendation: 70% training, 15% validation, 15% test split
- Application: All learning-based methods

3. Ablation Studies

- Purpose: Assess individual contributions of components
- Method: Systematically remove components and measure performance degradation
- Application: All multi-component methods (e.g., quality-aware NBV, integrated planning)

4. Sensitivity Analysis

- Purpose: Assess robustness to parameter variations
- Method: Vary parameters (e.g., ±10%, ±20%) and measure performance changes
- Application: All methods with tunable parameters

5. Independent Replication

- Purpose: Verify reproducibility of results
- Method: Independent researchers replicate experiments using published code and data
- Recommendation: Provide open-source code, datasets, and detailed documentation

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C.2.4 Experimental Design Recommendations

General Principles

1. Randomization

- Randomize trial order to control for learning effects and temporal confounds
- Use blocked randomization to ensure balanced conditions

2. Blinding

- Blind experimenters to condition assignments when possible
- Use automated data collection and analysis to minimize experimenter bias

3. Replication

- Conduct sufficient repeated trials ($n \geq 30$ per condition) for statistical power

- Replicate key experiments across multiple sessions and environments

4. Ground Truth

- Use independent measurement systems (laser trackers, photogrammetry) for ground truth
- Validate ground truth accuracy and precision

5. Documentation

- Document all experimental procedures, parameters, and deviations
- Provide detailed supplementary materials for reproducibility

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Platform-Specific Recommendations

1. Humanoid Platform (Unitree G1)

- Safety: Implement emergency stop procedures; use safety barriers
- Calibration: Calibrate camera intrinsics/extrinsics; validate kinematic model
- Stability: Monitor center of mass; implement fall detection and recovery
- Repeatability: Conduct repeated trials to assess pose repeatability

2. Manipulator Platform (Franka Panda)

- Safety: Implement collision detection; use safety-rated controllers
- Calibration: Calibrate camera intrinsics/extrinsics; validate kinematic model
- Repeatability: Conduct repeated trials to assess pose repeatability

3. Mobile Robot Platform (Clearpath Husky + UR5)

- Safety: Implement obstacle detection; use safety-rated controllers
- Calibration: Calibrate camera intrinsics/extrinsics; validate kinematic model
- Localization: Use independent localization system (motion capture, laser tracker)

C.2.5 Recommended Evaluation Metrics

Coverage Metrics

1. Coverage Completeness

- Definition: Percentage of target surface observed above resolution threshold
- Calculation: (Observed surface area) / (Total surface area) × 100%
- Threshold: Minimum resolution (e.g., 1 mm/pixel for defect detection)

2. Coverage Efficiency

- Definition: Coverage completeness per unit time or per viewpoint
- Calculation: (Coverage completeness) / (Execution time or number of viewpoints)

3. Coverage Redundancy

- Definition: Average number of times each surface point is observed
- Calculation: (Total observations) / (Unique surface points observed)

Quality Metrics

1. Acquisition Quality Score

- Definition: Composite metric combining blur, exposure, occlusion, framing
- Components:
  - Blur score: Laplacian variance or gradient magnitude
  - Exposure score: Histogram distribution or entropy
  - Occlusion score: Percentage of target visible
  - Framing score: Target size and centering in image
- Calculation: Weighted average of component scores (0–1 scale)

2. Re-Observation Frequency

- Definition: Percentage of viewpoints requiring re-observation due to poor quality
- Calculation: (Number of re-observations) / (Total viewpoints) × 100%

Repeatability Metrics

1. Pose Repeatability

- Definition: Standard deviation of camera pose across repeated trials
- Calculation: σ(position) in mm, σ(orientation) in degrees
- Measurement: Laser tracker or motion capture system

2. Image Attribute Consistency

- Definition: Standard deviation of image attributes across repeated trials
- Attributes: Resolution, blur, exposure, framing
- Calculation: σ(attribute) for each attribute

Efficiency Metrics

1. Execution Time

- Definition: Total time from start to completion
- Components: Planning time, motion time, acquisition time, quality assessment time

2. Motion Distance

- Definition: Total distance traveled by robot end-effector or base
- Calculation: Sum of Euclidean distances between consecutive poses

3. Number of Viewpoints

- Definition: Total number of viewpoints required for complete coverage
- Calculation: Count of unique camera poses

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